AI Project 4: Exam Preparation Solutions

Chat GPT-5 for DTL

1 Problem 1: Lubrication and Cavitation in Separating Plates

1.1 Restatement of the Problem

Two flat plates of length L meet at a vertex and form a very small opening angle. The local gap between the plates is

$$h(x,t) = \theta(t)x,$$

where x is the distance from the vertex and $\theta(t)$ is small and changing in time. The plates are being pried apart so that $\dot{\theta} = d\theta/dt > 0$. The liquid in the wedge is Newtonian with viscosity μ. Cavitation occurs when the *absolute* pressure in the liquid reaches zero (equivalently the gauge pressure reaches $-p_{\text{atm}}$).

We must find:

a. The location x_c where the liquid first cavitates, i.e. where $p = -p_{\text{atm}}$.

b. The force per unit width F/W at the outer edge $x = L$ required to pry the plates apart at angular velocity $\dot{\theta}$.

1.2 Solution

1.2.1 Lubrication equation

Under lubrication assumptions, the 1-D Reynolds equation for squeezing flow is

$$\frac{d}{dx}\left(-\frac{h^3}{12\mu}\frac{dp}{dx}\right) = -\frac{dh}{dt},$$

or equivalently

$$\frac{d}{dx}\left(h^3\frac{dp}{dx}\right) = 12\mu\frac{dh}{dt}.$$

For $h(x,t) = \theta(t)x$, we have

$$h^3 = \theta^3 x^3, \qquad \frac{dh}{dt} = x\dot{\theta}.$$

Substitute:

$$\frac{d}{dx}\left(\theta^3 x^3\frac{dp}{dx}\right) = 12\mu x\dot{\theta}.$$

Integrate once with respect to x:

$$\theta^3 x^3\frac{dp}{dx} = 6\mu\dot{\theta}x^2 + C_1.$$

Regularity at $x = 0$ requires $C_1 = 0$, hence

$$\frac{dp}{dx} = \frac{6\mu\dot{\theta}}{\theta^3}\frac{1}{x}.$$

Integrate again:

$$p(x) = \frac{6\mu\dot{\theta}}{\theta^3} \ln x + C_2.$$

Apply $p(L) = 0$:

$$C_2 = -\frac{6\mu\dot{\theta}}{\theta^3} \ln L.$$

Thus the pressure distribution (gauge) is

$$p(x) = K \ln\left(\frac{x}{L}\right), \qquad K = \frac{6\mu\dot{\theta}}{\theta^3}. \tag{1}$$

1.2.2 (a) Location of cavitation

Cavitation occurs where $p(x_c) = -p_{\text{atm}}$. From (1),

$$K \ln\left(\frac{x_c}{L}\right) = -p_{\text{atm}},$$

so

$$x_c = L \exp\left(-\frac{p_{\text{atm}}\theta^3}{6\mu\dot{\theta}}\right).$$

1.2.3 (b) Force per unit width

Take moments about the vertex. The fluid exerts a resisting moment

$$M_{\text{fluid}} = \int p(x) x \, dx$$

over the wetted region.

No cavitation (wetted from 0 to L):

$$M_{\text{fluid}} = \int_0^L K \ln\left(\frac{x}{L}\right) x \, dx.$$

Let $u = x/L$:

$$M_{\text{fluid}} = KL^2 \int_0^1 u \ln u \, du = -\frac{1}{4} KL^2.$$

Moment balance: $(F/W)L = M_{\text{fluid}}$, hence

$$\frac{F}{W} = -\frac{KL}{4} = -\frac{3}{2} \frac{\mu L\dot{\theta}}{\theta^3}.$$

The magnitude of the applied force per unit width is

$$\left|\frac{F}{W}\right| = \frac{3}{2} \frac{\mu L\dot{\theta}}{\theta^3}.$$

1.3 Summary of Key Formulae

$$
\begin{aligned}
K &= \frac{6\mu\dot{\theta}}{\theta^3}, \\
p(x) &= K \ln\left(\frac{x}{L}\right), \\
x_c &= L \exp\left(-\frac{p_{\mathrm{atm}}\theta^3}{6\mu\dot{\theta}}\right), \\
\frac{F}{W} &= -\frac{3}{2}\frac{\mu L\dot{\theta}}{\theta^3},
\end{aligned}
$$

1.4 Study Guide

Key Concepts

1. **Lubrication Approximation and Reynolds Equation.** Derivation of $d(h^3 dp/dx)/dx = 12\mu dh/dt$ for thin films; connection between pressure gradients and film deformation.

2. **Boundary Conditions and Regularity.** Pressure at outer edge equals ambient; singularity avoidance at the vertex enforces $C_1 = 0$.

3. **Cavitation Criterion and Force Balance.** Gauge pressure limited by $-p_{\mathrm{atm}}$; integrate $p(x)x$ for torque/force evaluation.

Most Applicable Lecture This problem corresponds to
Lecture 15: Lubrication Flows at `https://diffusivepress.org/courseware/cbe30355/notes/l15/notes.html` and
Lecture 16: The Reynolds Lubrication Equation at `https://diffusivepress.org/courseware/cbe30355/notes/l16/notes.html`.

Practice Variation (Unsolved)

> Consider the same wedge geometry $h(x,t) = \theta(t)x$ but with the lower plate moving horizontally at velocity U while $\dot{\theta} > 0$. Derive the modified pressure distribution $p(x)$ including both squeeze and Couette effects, and determine how the cavitation point x_c changes with U.

1.5 Notes

1. The above analysis uses the Reynolds lubrication equation to calculate the pressure gradient. An alternative approach assuming a defined flow rate (resulting from conservation of volume in the expanding wedge) yields the same result.

2. The analysis assumes regularity of the pressure at the origin. In fact, however, cavitation means that the expression is not valid for $x < x_c$, and thus the proper boundary condition requires that instead $p(x_c) = -p_{atm}$. This leads to a much more complicated expression where the net flow in the cavity must account for the transient evolution of the location of cavitation. This effect is not considered here.

1.6 Video Narration

The video narration may be found at: `https://notredame.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=40585686-cba3-496c-8ace-b39f0143670b`

2 Inertial Impaction and Face Shields

2.1 Restatement of the problem

A spherical droplet of radius a moves with the ambient fluid at speed U_0. At time $t = 0$ the fluid velocity is instantaneously stopped (representing deflection by a rigid face shield). Because of its inertia the droplet continues to move and may impact the shield. We are asked to:

(a) Use dimensional analysis to determine the dimensionless groups on which the droplet displacement x depends.

(b) In the Stokes (low Reynolds number) limit, use the linearity of the governing equations to reduce the dimensional analysis to a single group.

(c) Solve the time-dependent motion of the particle using Newton's second law with Stokes drag and obtain the velocity and displacement as functions of time.

(d) For $U_0 = 100$ cm s^{-1}, fluid viscosity $\mu = 1.81 \times 10^{-4}$ g cm^{-1}s^{-1}, distance to the shield $L = 5$ cm, estimate the droplet diameter that will be captured by inertial impaction (i.e., satisfy $x \geq L$).

2.2 Solution

2.2.1 (a) Dimensional analysis

List the relevant physical variables and their dimensions (in mass M, length L, time T):

$$x\ (L), \quad a\ (L), \quad \rho\ (ML^{-3}), \quad \rho_a\ (ML^{-3}), \quad \mu\ (ML^{-1}T^{-1}), \quad U_0\ (LT^{-1}).$$

We have 6 variables and 3 fundamental dimensions M, L, T, so by Buckingham Π theorem there are $6 - 3 = 3$ independent dimensionless groups. A convenient choice is:

$$\Pi_1 = \frac{x}{a} \quad \text{(dimensionless displacement)},$$

$$\Pi_2 = \frac{\rho}{\rho_a} \quad \text{(particle-to-air density ratio)},$$

$$\Pi_3 = \mathrm{Re} = \frac{\rho_a U_0 a}{\mu} \quad \text{(Reynolds number based on air and particle size)}.$$

Thus in full generality,

$$\frac{x}{a} = F\left(\frac{\rho}{\rho_a}, \frac{\rho_a U_0 a}{\mu}\right).$$

2.2.2 (b) Stokes (low-Re) limit and reduction

If the droplet motion occurs at very small particle Reynolds number (so that viscous Stokes drag applies and the flow dynamics are linear in velocity) and the inertial effects of the *air* itself are negligible, the only fluid density that remains relevant is the particle density ρ (air density ρ_a may be neglected from the force balance). Linearity implies the final displacement x scales linearly with the initial velocity U_0, so we can write

$$x = U_0\, \tau(\rho, \mu, a),$$

where τ has dimensions of time and depends on ρ, μ, a. From dimensions one finds a single combination for τ made from ρ, μ, a:

$$\tau \sim \frac{\rho a^2}{\mu}.$$

Hence a single reduced dimensionless group may be written as

$$\frac{x}{U_0} \cdot \frac{\mu}{\rho a^2} = \text{constant (order 1)}.$$

Equivalently,

$$\frac{x}{a} = C\, \frac{\rho U_0 a}{\mu},$$

for some constant C of order unity determined by the detailed solution.

2.2.3 (c) ODE solution (Newton + Stokes drag)

We now set up the precise differential equation and solve for velocity and displacement.

Mass of the spherical droplet:

$$m = \frac{4}{3}\pi a^3 \rho.$$

Stokes drag on a sphere moving through a quiescent fluid (for small particle Reynolds number) is

$$F_D = -6\pi\mu a\, u(t),$$

where $u(t)$ is the particle velocity relative to the (now zero) fluid.

Newton's second law gives

$$m\frac{du}{dt} = -6\pi\mu a\, u.$$

This is a linear first-order ODE. Define the relaxation time (decay time)

$$\tau \equiv \frac{m}{6\pi\mu a} = \frac{\frac{4}{3}\pi a^3 \rho}{6\pi\mu a} = \frac{2\rho a^2}{9\mu}.$$

The ODE becomes

$$\frac{du}{dt} = -\frac{1}{\tau}u, \qquad u(0) = U_0.$$

Solve:

$$u(t) = U_0\, e^{-t/\tau}.$$

The displacement measured from $t = 0$ is

$$x(t) = \int_0^t u(t')\, dt' = U_0\tau\left(1 - e^{-t/\tau}\right).$$

The total eventual displacement as $t \to \infty$ is

$$x_\infty = U_0\tau = U_0\, \frac{2\rho a^2}{9\mu}.$$

Thus the constant C from part (b) is precisely $2/9$.

2.2.4 (d) Numerical estimate for capture by inertial impaction

Inertial impaction occurs if the total displacement x_∞ exceeds the distance to the shield, L. Thus require

$$x_\infty = U_0 \frac{2\rho a^2}{9\mu} \geq L.$$

Solve for the particle radius a (taking ρ of the droplet as that of liquid water, $\rho \approx 1$ g cm^{-3}):

$$a \geq \sqrt{\frac{9\mu L}{2\rho U_0}}.$$

Insert the given numeric values (all in CGS units):

$$U_0 = 100 \text{ cm}\,\text{s}^{-1}, \quad \mu = 1.81 \times 10^{-4} \text{ g}\,\text{cm}^{-1}\,\text{s}^{-1}, \quad L = 5 \text{ cm}, \quad \rho = 1.0 \text{ g}\,\text{cm}^{-3}.$$

Compute:

$$a = \sqrt{\frac{9(1.81 \times 10^{-4})(5)}{2(1.0)(100)}} = 6.38 \times 10^{-3} \text{ cm}.$$

Thus the corresponding diameter $d = 2a$ is

$$d \approx 1.28 \times 10^{-2} \text{ cm} = 128 \ \mu\text{m}.$$

Interpretation: droplets with diameter on the order of $\sim 1.3 \times 10^2$ μm (about 0.13 mm) or larger will be stopped by inertial impaction given the stated parameters. Much smaller droplets will travel past the shield because their viscous drag removes their inertia before they cover the required distance.

2.3 Study guide

1. **Key concepts to master**
   (a) Buckingham Π theorem and formation of dimensionless groups (how to choose repeating variables and interpret groups like Re and density ratios).
   (b) Stokes drag on a sphere: $F_D = 6\pi\mu a u$ (valid for very small particle Reynolds numbers) and its use in force balances for particles.
   (c) Solving first-order linear ODEs for exponential relaxation; interpretation of the relaxation time τ and the relation $x_\infty = U_0\tau$.
2. **Most applicable lecture(s)**:
   Lecture 14: Dimensional Analysis located at `https://diffusivepress.org/courseware/cbe30355/notes/l14/notes.html`
   Lecture 18: Flow Past a Sphere located at `https://diffusivepress.org/courseware/cbe30355/notes/l18/notes.html`.
3. **Additional practice problem (do not solve here)**:
   Variation: Consider a spherical droplet of radius a that initially moves at speed U_0 in a fluid which for $t > 0$ has a uniform constant backward flow U_f (instead of zero). Using Stokes drag and Newton's second law, determine the particle velocity $u(t)$ and the steady-state displacement relative to the fluid as $t \to \infty$. Express your answer in terms of U_0, U_f, ρ, μ, a.

2.4 Short note on assumptions and limitations

- The analytical solution used Stokes drag and neglected added-mass and unsteady fluid inertial forces (Basset history term and added mass). These effects can be important for larger Reynolds numbers or rapid accelerations and would modify the effective relaxation time. The Stokes approximation is valid when the particle Reynolds number $\mathrm{Re}_p = \rho_a U_0 a/\mu \ll 1$.

- We assumed the droplet density equals that of liquid water. For other materials or evaporating droplets the density (and radius) may change.

- The model is one-dimensional and neglects the detailed 3D flow deflection geometry around a real face shield; however it captures the essential inertial-impaction mechanism.

2.5 Video Narration

The video narration may be found at: `https://notredame.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=24e50efd-b28d-455d-b27b-b39a011c4e40`

3 Problem 3: Capillary Filling of a Tube by Surface Tension

3.1 Restatement of the Problem

A liquid (blood) is drawn into a horizontal cylindrical capillary of radius a by surface tension. The surface tension is $\Gamma = 56$ dynes/cm, the dynamic viscosity is $\mu = 4$ cP, the capillary radius is $a = 50\ \mu$m, and the tube length is $L = 5$ cm. Inertial effects are negligible and the capillary pressure $\Delta p_{cap} = 2\Gamma/a$ acts across the filled length $h(t)$. Determine how long it takes for the tube to fill completely.

3.2 Solution

3.2.1 a. Governing equations and boundary conditions

We start from the axisymmetric incompressible Navier–Stokes equations in cylindrical coordinates (given in the problem statement). Under the assumptions:

- steady viscous-dominated (negligible inertia) flow in the filled portion,
- fully developed axisymmetric Poiseuille profile in the axial direction over the instantaneous filled length $0 \leq z \leq h(t)$,
- no slip at the wall $v_z(r = a) = 0$ and finite v_z at $r = 0$,
- pressure at the meniscus is lower than the reservoir by the capillary pressure: $p_{meniscus} = p_\infty - \Delta p_{cap}$ where $\Delta p_{cap} = 2\Gamma/a$,
- pressure at the reservoir (outside the tube) is p_∞.

In the viscous-dominated limit the axial momentum equation reduces to the Stokes equation for axial velocity $v_z(r, z)$:

$$0 = -\frac{\partial p}{\partial z} + \mu \left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right)\right],$$

where we have assumed negligible dependence on θ and negligible axial diffusion term for fully developed flow in z (so $v_z = v_z(r)$ for a given axial pressure gradient).

For a constant (in r) axial pressure gradient dp/dz over the filled length $0 \leq z \leq h(t)$, the Poiseuille solution for the volumetric flow rate Q is

$$Q = -\frac{\pi a^4}{8\mu}\frac{\Delta p}{h},$$

where $\Delta p = p_\infty - p_{meniscus} = \Delta p_{cap} = 2\Gamma/a$ and the effective pressure drop is applied over length h. The mean (average) axial velocity is

$$\bar{u} = \frac{Q}{\pi a^2} = -\frac{a^2}{8\mu}\frac{\Delta p}{h}.$$

The kinematic relation between the meniscus position $h(t)$ and the mean velocity is

$$\frac{dh}{dt} = \bar{u}.$$

Taking magnitudes (flow into the tube increases h):

$$\frac{dh}{dt} = \frac{a^2}{8\mu}\frac{\Delta p_{cap}}{h} = \frac{a^2}{8\mu}\frac{2\Gamma/a}{h} = \frac{a\Gamma}{4\mu}\frac{1}{h}.$$

This is the governing ordinary differential equation (ODE) for $h(t)$ with initial condition $h(0) = 0$.

b. Non-dimensionalization

Define dimensionless variables
$$h^* = \frac{h}{L}, \quad t^* = \frac{t}{t_c},$$
where we choose the characteristic time scale t_c so that the dimensionless ODE has simple coefficients. Using the ODE
$$\frac{dh}{dt} = \frac{a\Gamma}{4\mu}\frac{1}{h},$$
substitute $h = Lh^*$ and $t = t_c t^*$:
$$\frac{L}{t_c}\frac{dh^*}{dt^*} = \frac{a\Gamma}{4\mu}\frac{1}{Lh^*}.$$
Rearrange:
$$\frac{dh^*}{dt^*} = \left(\frac{t_c a\Gamma}{4\mu L^2}\right)\frac{1}{h^*}.$$
Choose
$$t_c = \frac{2\mu L^2}{a\Gamma},$$
which yields
$$\frac{dh^*}{dt^*} = \frac{1}{2h^*}.$$
Multiplying both sides by h^* gives
$$h^*\frac{dh^*}{dt^*} = \frac{1}{2},$$
or
$$\frac{d}{dt^*}\left(\frac{1}{2}h^{*2}\right) = \frac{1}{2}.$$
Thus the dimensionless solution will satisfy
$$h^{*2} = t^* + C^*,$$
with $C^* = 0$ from $h^*(0) = 0$. Hence
$$h^{*2} = t^*.$$
This shows the filling time scales as
$$t_c = \frac{2\mu L^2}{a\Gamma},$$
so the dimensional filling time for $h = L$ is proportional to $\mu L^2/(a\Gamma)$.

c. Solution for the filling time

Integrate the ODE directly. From
$$h\frac{dh}{dt} = \frac{a\Gamma}{4\mu},$$
we have
$$\frac{1}{2}\frac{d}{dt}(h^2) = \frac{a\Gamma}{4\mu}.$$

Integrating from $t = 0$ (with $h(0) = 0$) to time t gives

$$\frac{1}{2}h^2 = \frac{a\Gamma}{4\mu}\,t.$$

Hence

$$h^2 = \frac{a\Gamma}{2\mu}\,t, \quad \text{or} \quad h(t) = \sqrt{\frac{a\Gamma}{2\mu}\,t}.$$

The time to fill the tube to length L (i.e., $h = L$) is

$$t_{fill} = \frac{2\mu L^2}{a\Gamma}.$$

3.2.2 d. Numerical evaluation

First convert given quantities to SI units:

$$\begin{aligned}
&\Gamma = 56 \text{ dynes/cm} = 56 \times 10^{-3} \text{ N/m} = 0.056 \text{ N/m},\\
&\mu = 4 \text{ cP} = 4 \times 10^{-3} \text{ Pa s} = 0.004 \text{ Pa s},\\
&a = 50\ \mu\text{m} = 50 \times 10^{-6} \text{ m} = 5.0 \times 10^{-5} \text{ m},\\
&L = 5 \text{ cm} = 0.05 \text{ m}.
\end{aligned}$$

Compute

$$t_{fill} = \frac{2\mu L^2}{a\Gamma}.$$

Substituting values:

$$t_{fill} = \frac{2(0.004)(0.05)^2}{(5.0 \times 10^{-5})(0.056)}.$$

Evaluate numerator and denominator:

$$\text{numerator} = 2 \times 0.004 \times 0.0025 = 2.0 \times 10^{-5}, \quad \text{denominator} = 5.0 \times 10^{-5} \times 0.056 = 2.8 \times 10^{-6}.$$

Hence

$$t_{fill} = \frac{2.0 \times 10^{-5}}{2.8 \times 10^{-6}} \approx 7.14 \text{ s}.$$

Thus the capillary fills in approximately $t_{fill} \approx 7.1$ seconds.

3.3 Study Guide

3.3.1 Key concepts to master

1. **Capillary pressure and surface tension:** A curved liquid meniscus creates a pressure difference $\Delta p_{cap} = 2\Gamma/a$ for a cylindrical tube (assuming complete wetting), which drives the flow.
2. **Poiseuille flow and viscous resistance:** The volumetric flow rate for laminar flow in a pipe is $Q = -(\pi a^4/8\mu)(\Delta p/h)$, and the relation between average velocity and flow rate is $\bar{u} = Q/(\pi a^2)$.
3. **Scaling and non-dimensionalization:** Choose characteristic scales so that $t_c = 2\mu L^2/(a\Gamma)$; this clarifies the parameter dependence of filling time on μ, L, a, and Γ.

3.3.2 Most applicable lecture

This problem is most closely related **Lecture 11: Poiseuille Flow & Couette Flow** located at `https://diffusivepress.org/courseware/cbe30355/notes/l11/notes.html`.

3.3.3 Additional practice problem (do not solve here)

> A vertical capillary tube of radius a is partially immersed in a reservoir of liquid with density ρ, viscosity μ, and surface tension Γ. The contact angle is θ (not necessarily zero). Derive the time-dependent rise $h(t)$ of the liquid column accounting for both capillary pressure and hydrostatic pressure (i.e., the driving pressure is $\Delta p_{cap} - \rho g h$). Assume viscous-dominated flow and neglect inertia. Determine the equilibrium height and the approach rate to that equilibrium.

3.4 Notes and assumptions

- We assumed complete wetting ($\theta \approx 0$) so the capillary pressure used is $2\Gamma/a$. If θ is not zero, replace Γ by $\Gamma \cos\theta$.
- Inertial effects were neglected as stated; for very short times or low viscosity, these may become important.
- The derived Washburn-type relation $h \propto \sqrt{t}$ is the classic form for capillary imbibition under viscous-dominated flow.

3.5 Video Narration

The video narration may be found at: `https://notredame.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=f2b35412-4cc6-427a-a956-b39a011c4e43`

4 Problem 4: Spin Coating and Thin Film Dynamics

4.1 Restatement of Problem

A viscous liquid film of kinematic viscosity ν coats a rotating disk of radius R spinning at angular velocity Ω. The centrifugal force drives the liquid radially outward, thinning the film over time until a final thickness $h_f \ll R$ is reached. We are asked to:

- **a.** Derive the governing equations and boundary conditions for the velocity field and film thickness evolution.
- **b.** Estimate the characteristic time for thinning using scaling arguments.
- **c.** Solve explicitly for $u_r(r, z, t)$ and $h(t)$, determining the $\mathcal{O}(1)$ constant in the time scale.

4.2 Solution

4.2.1 a. Governing Equations and Boundary Conditions

We assume an axisymmetric thin film ($h \ll R$) and negligible inertia in z. The continuity and momentum equations simplify to:

$$\frac{1}{r}\frac{\partial(ru_r)}{\partial r} + \frac{\partial u_z}{\partial z} = 0, \tag{2}$$

$$0 = -\frac{\partial p}{\partial r} + \mu\frac{\partial^2 u_r}{\partial z^2} + \rho\Omega^2 r. \tag{3}$$

Boundary conditions:

$$\begin{aligned}
u_r(r,0,t) &= 0, && \text{(no slip at disk)}\\
\frac{\partial u_r}{\partial z}(r,h,t) &= 0, && \text{(zero shear at free surface)}\\
u_z(r,0,t) &= 0, && \text{(no penetration)}\\
\frac{\partial p}{\partial z} &= 0, && \text{(negligible curvature and gravity in } z\text{).}
\end{aligned}$$

Integrating Eq. (3) twice with respect to z:

$$u_r = \frac{\rho\Omega^2 r}{2\mu}(z^2 - 2hz),$$

using the boundary conditions above.

The mean (depth-averaged) radial velocity is:

$$\bar{u}_r = \frac{1}{h}\int_0^h u_r\, dz = -\frac{\rho\Omega^2 r h^2}{3\mu}.$$

From continuity (2), the kinematic condition at the free surface gives:

$$\frac{\partial h}{\partial t} + \frac{1}{r}\frac{\partial}{\partial r}(rh\bar{u}_r) = 0.$$

Substituting for $\bar{u}_r$:

$$\frac{\partial h}{\partial t} = \frac{\rho\Omega^2}{3\mu r}\frac{\partial}{\partial r}\left(r^2 h^3\right).$$

For uniform thinning ($\partial h/\partial r \approx 0$) near the center, the dominant term is:

$$\frac{dh}{dt} \approx -\frac{2\rho\Omega^2 h^3}{3\mu}.$$

4.2.2 b. Scaling Analysis

Let t_s be the characteristic thinning time. Balancing the terms:

$$\frac{h}{t_s} \sim \frac{\rho\Omega^2 R^2 h^3}{\mu R^2} \Rightarrow t_s \sim \frac{\mu}{\rho\Omega^2 h^2}.$$

The process duration depends inversely on Ω^2 and on the square of the film thickness. Using $h \sim h_f$ at late times, we estimate:

$$t_s \sim \frac{\mu}{\rho\Omega^2 h_f^2}.$$

4.2.3 c. Explicit Solution

From

$$\frac{dh}{dt} = -\frac{2\rho\Omega^2}{3\mu} h^3,$$

we integrate from $h(0) = H$ to $h(t)$:

$$\int_H^{h(t)} h^{-3} dh = -\frac{2\rho\Omega^2}{3\mu} \int_0^t dt,$$

yielding

$$\frac{1}{2}\left(\frac{1}{h^2} - \frac{1}{H^2}\right) = \frac{2\rho\Omega^2 t}{3\mu}.$$

Thus,

$$h(t) = \frac{H}{\sqrt{1 + \dfrac{4\rho\Omega^2 H^2 t}{3\mu}}}.$$

The $\mathcal{O}(1)$ constant in the time scale is therefore 4/3.

4.2.4 Final Thickness and Time Scale

If the final thickness is $h_f \ll H$, the time to reach it is

$$t_f = \frac{3\mu}{4\rho\Omega^2}\left(\frac{1}{h_f^2} - \frac{1}{H^2}\right) \approx \frac{3\mu}{4\rho\Omega^2 h_f^2}.$$

4.3 Study Guide

4.3.1 Key Concepts to Master

1. **Lubrication approximation:** Recognizing that for thin films ($h \ll R$), vertical velocity gradients dominate, simplifying the Navier–Stokes equations.
2. **Centrifugal-driven flow:** Understanding how $\rho\Omega^2 r$ acts as a body force in the radial direction.
3. **Kinematic boundary condition:** Knowing how $\partial h/\partial t$ is related to the vertical velocity at the free surface.

4.3.2 Most Applicable Lectures

This problem contains elements from

1. **Lecture 10: Flow Down an Incline** located at: `https://diffusivepress.org/courseware/cbe30355/notes/l10/notes.html`
2. **Lecture 12: Couette Flow** (for a discussion of centrifugal force) located at: `https://diffusivepress.org/courseware/cbe30355/notes/l12/notes.html`.
3. **Lecture 15: Lubrication Flows** located at: `https://diffusivepress.org/courseware/cbe30355/notes/l15/notes.html`.

4.3.3 Additional Practice Problem

A volatile solvent is spin-coated on a rotating disk of angular velocity Ω. In addition to centrifugal thinning, evaporation removes liquid from the surface at a constant rate E (m/s). Derive the modified film-thickness evolution equation and find the steady-state film thickness if one exists.

4.4 Video Narration

The video narration may be found at: `https://notredame.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=4d52cdc0-6dc4-4d21-bbc8-b39a011c4e41`